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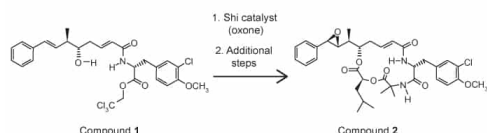


Figure 1 Shi epoxidation of Compound 1

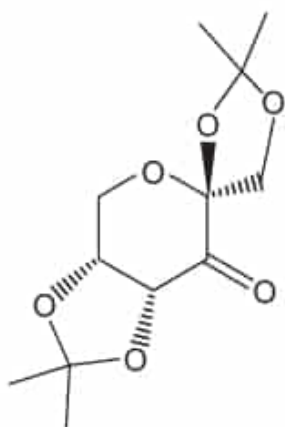


Figure 2 Structure of the Shi catalyst

The epoxidation of Compound 1 was monitored by thin-layer chromatography (TLC). When the reaction was completed, high-performance liquid chromatography (HPLC) was used to analyze the epoxide product. A chiral column was used that had a higher affinity for the desired epoxide than for other isomers. A standard of the desired epoxide was obtained to optimize the conditions and determine the retention time of the epoxide. The HPLC chromatogram of the epoxide standard is shown in Figure 3.

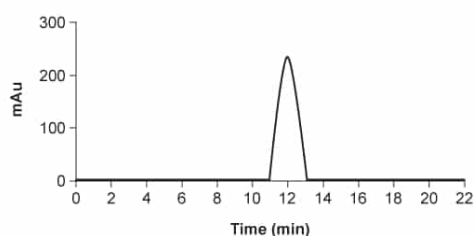


Figure 3 HPLC of epoxide standard

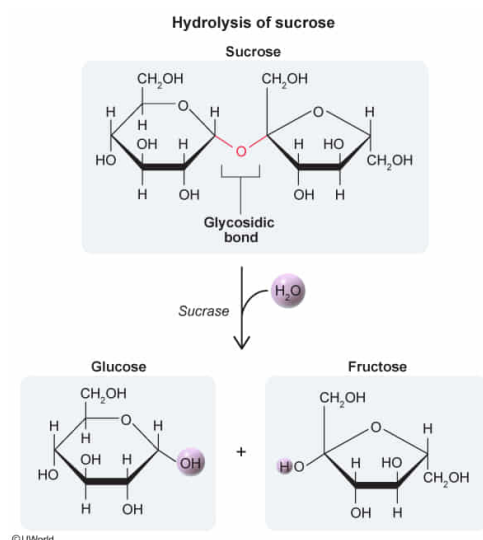
Fructose for the synthesis of the Shi catalyst can be obtained by cleaving sucrose. What type of reaction occurs when sucrose is cleaved?

- ☐ A. Condensation of a glycosidic bond [6%]
☐ B. Condensation of a peptide bond [1%]
☐ C. Hydrolysis of a peptide bond [2%]
☒ D. Hydrolysis of a glycosidic bond [89%]

Correct

89% answered correctly

Explanation:



A **glycosidic bond** is the **linkage** between two **sugar molecules** in a disaccharide or polysaccharide, or between a sugar molecule and a hydroxyl group of another molecule. This bond is made up of the **hemiacetal** or **hemiketal** of one sugar and the **hydroxyl (–OH) group** of another connected via an **α- or β-linkage**. A **glycosidic bond** is broken via a **hydrolysis reaction**, frequently with the aid of an enzyme.

In this reaction, water acts as a **nucleophile** and breaks the glycosidic bond, with one of the sugars acting as a leaving group. Therefore, the sugar is broken into two smaller components.

The disaccharide sucrose is made up of fructose and glucose joined together by a glycosidic bond. To obtain fructose from sucrose, the glycosidic bond must be hydrolyzed.

(Choices A and B) A sugar is cleaved during a reaction known as hydrolysis instead of during condensation. A **condensation reaction** is the opposite of hydrolysis, requiring the loss of water as two units are joined together. In the case of carbohydrates, this process forms a glycosidic bond rather than a **peptide bond**.

(Choice C) Hydrolysis is the reaction necessary to obtain fructose from sucrose. However, peptide bonds are not found in sugars but rather link **amino acids** together in proteins.

Educational objective:

A glycosidic bond is the linkage between monosaccharides and is made up of a hemiacetal or hemiketal from one sugar and the hydroxyl group of another molecule. The glycosidic bond is broken through a hydrolysis reaction in which water cleaves the bond, splitting the sugar molecule.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401152

System:
Functional Groups and Their Reactions

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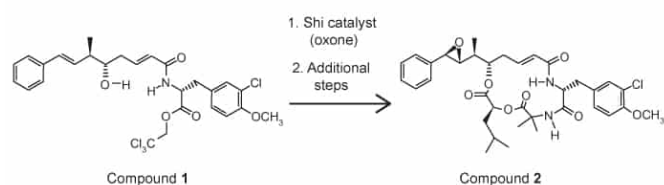


Figure 1 Shi epoxidation of Compound 1

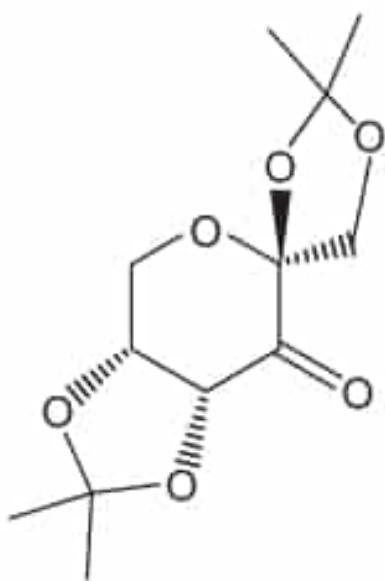


Figure 2 Structure of the Shi catalyst

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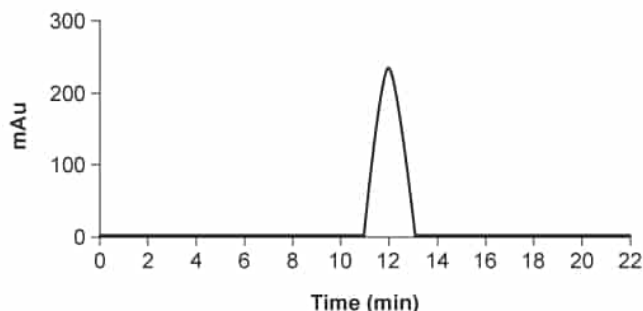
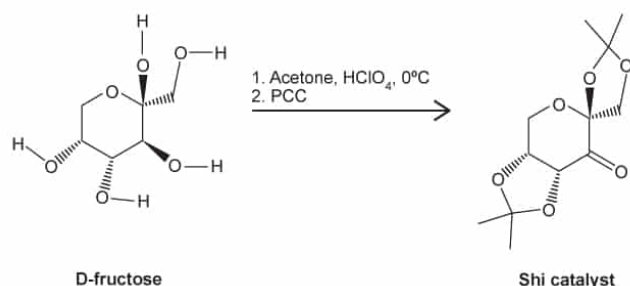


Figure 3 HPLC of epoxide standard

The Shi catalyst is derived from D-fructose based on the reaction below.

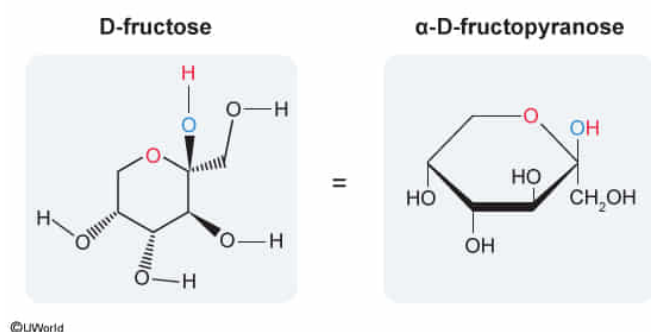


The structure of D-fructose as shown in the reaction can be classified as:

- I. pyranose
- II. furanose
- III. ketohexose
- IV. aldohexose

- ✓ ☒ A. I and III only [48%]
- ☐ B. I and IV only [23%]
- ☐ C. II and III only [18%]
- ☐ D. II and IV only [9%]

Explanation:



Sugars are organic molecules composed of a carbon chain in which one carbon is double-bonded to oxygen and the rest contain a hydroxyl group. They are classified based on several characteristics: the number of carbon atoms they contain, the functional groups in the open chain form, and what type of cyclic structure (if any) they have adopted.

Sugar classes are named first for the number of carbon atoms they contain by adding the **suffix –ose** to the **Greek prefix** corresponding to the carbon number. For example, a five-carbon sugar is a pentose and a six-carbon sugar is a hexose. The sugar can be further classified as either an aldose or a ketose. An **aldose** is a sugar that contains an aldehyde functional group with the carbonyl at carbon 1, and a **ketose** contains a ketone, typically at carbon 2. **Fructose** is a six-carbon sugar with a ketone group, making it a **ketohehexose (Number III)**.

Some sugars, most commonly pentoses and hexoses, can take on cyclical structures known as pyranoses and furanoses. Pyranoses are six-membered rings and are named due to their similarity to **pyran**. Furanoses are five-membered rings and are named due to their resemblance to **furan**. Although fructose more commonly adopts the furanose form, it can adopt **either form**. The reaction diagram in the question shows that the form of D-fructose used to produce the Shi catalyst is a six-membered ring. Therefore, this form of fructose can be classified as a **pyranose (Number I)**.

(Number II) Although fructose is most commonly found in the furanose form, the diagram shows the pyranose form of fructose, not the furanose form.

(Number IV) Aldohexoses are sugars that contain six carbon atoms in the sugar backbone as well as an aldehyde functional group. **Glucose** is an

example of an aldohexose.

Educational objective:

Sugars can be classified by the number of carbon atoms in their backbone, whether they contain an aldehyde or a ketone, and by the cyclic form they adopt. The suffix –ose is used to name sugars.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401153

System:
Functional Groups and Their Reactions

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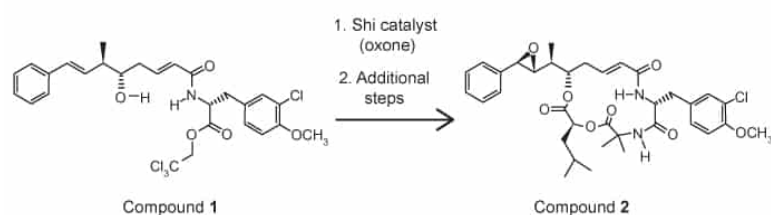


Figure 1 Shi epoxidation of Compound 1

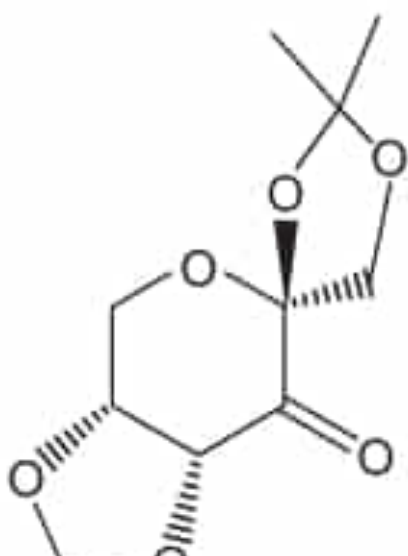




Figure 2 Structure of the Shi catalyst

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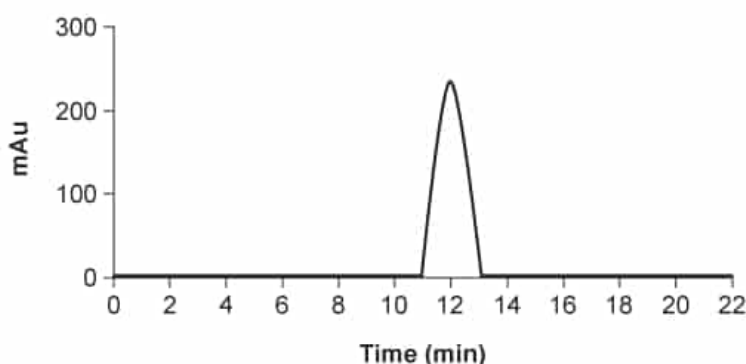


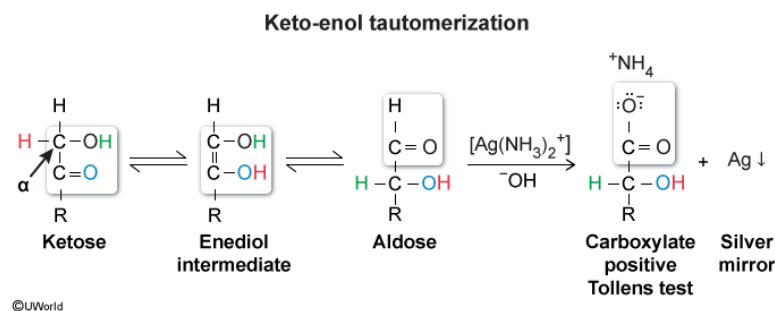
Figure 3 HPLC of epoxide standard

Ketoses such as fructose are expected to give a positive Tollens test because:

- ☐ A. ketoses cannot mutarotate. [8%]
- ☐ B. ketoses are not reducing sugars. [32%]
- ✓ ☒ C. ketoses tautomerize to aldoses. [43%]
- ☐ D. ketoses are not hemiacetals. [16%]

Correct

Explanation:



The **Tollens test** is used to identify the presence of **aldehydes** and **hydroxy ketones**, including reducing sugars, which have a free **anomeric carbon**. It utilizes the oxidizing agent $\text{Ag}(\text{NH}_3)_2^+$ to oxidize aldehydes to carboxylic acids. A positive Tollens test results in the formation of metallic silver that looks like a **silver mirror** deposited on the glassware.

Ketoses such as fructose are hydroxy ketones. Ketones cannot be directly oxidized to carboxylic acids; however, hydroxy ketones can undergo **tautomerization** via an enediol intermediate to produce an aldehyde under Tollens test basic conditions. Because of this tautomerization, ketoses give a positive result to a Tollens test.

(Choice A) **Mutarotation** involves the change in configuration (α and β) in sugars about the anomeric carbon in which the hemiacetal opens up to its linear form and then cyclizes back to a hemiacetal in the opposite configuration. Fructose is capable of mutarotation.

(Choice B) Any sugar with a free anomeric carbon (ie, an anomeric carbon that is not involved in a glycosidic bond) is a reducing sugar. Fructose has a free anomeric carbon, so it is a reducing sugar, not a nonreducing sugar.

(Choice D) Ketoses are a type of **hemiacetal**, a molecule in which a central carbon is part of both an ether and an alcohol. More specifically, ketoses are **hemiketals**, which are simply hemiacetals in which the central carbon is bonded to two other carbons instead of one carbon and one hydrogen. Therefore, fructose can exist in a hemiacetal form.

Educational objective:

The Tollens test is used to identify the presence of aldehydes and hydroxy ketones, and uses the oxidizing agent $[\text{Ag}(\text{NH}_3)_2]^+$ to oxidize aldehydes to carboxylic acids. Ketoses can undergo tautomerization via an enediol intermediate to their aldose form, resulting in a positive Tollens test and the formation of a silver mirror appearance.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401154

System:

Functional Groups and Their Reactions

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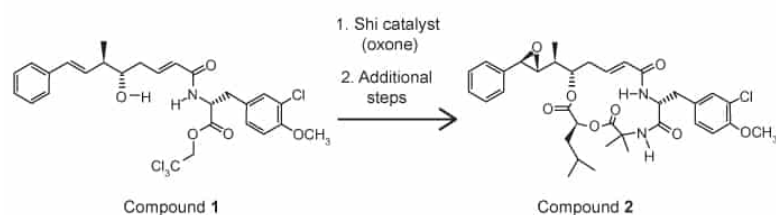


Figure 1 Shi epoxidation of Compound 1

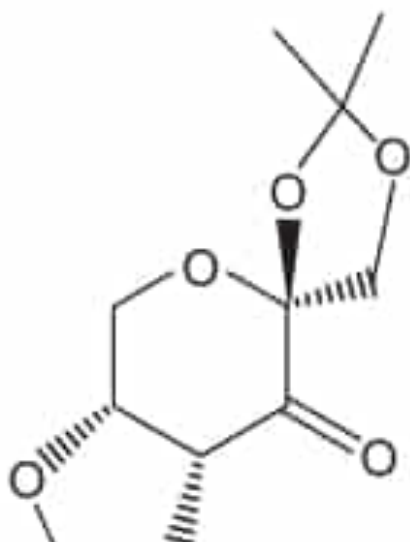




Figure 2 Structure of the Shi catalyst

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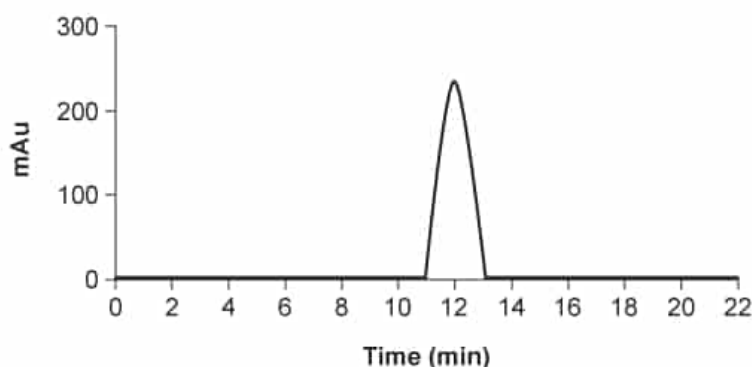


Figure 3 HPLC of epoxide standard

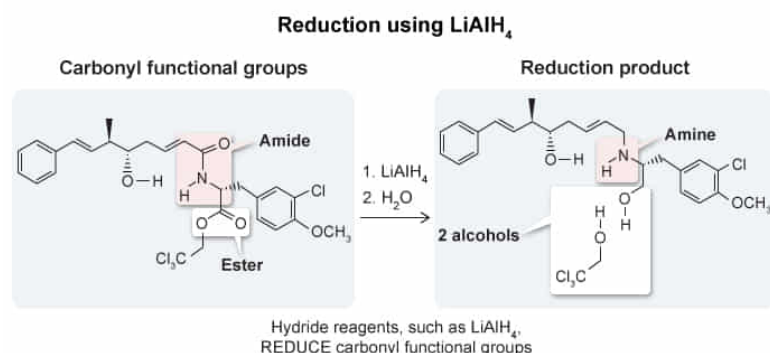
Which of the following functional groups will be reduced when LiAlH_4 is added to Compound **1**, the substrate for the Shi epoxidation described in the passage?

- ☐ A. Carbon-carbon double bond [11%]
- ✓ ☒ B. Ester [68%]
- ☐ C. Alcohol [18%]
- ☐ D. Aromatic ring

Correct

68% answered correctly

Explanation:



Reducing agents cause a decrease in an atom's **oxidation state**.

Reduction of carbon implies a **decrease** in the number of **bonds to electronegative atoms** (such as oxygen, nitrogen, or a halogen) and an **increase** in the number of **bonds to less electronegative atoms**, usually hydrogen. Metal hydride—reducing agents, such as LiAlH_4 , are made up of a metal surrounded by one or more H^- ions (hydride ions). The hydride ion acts as a **nucleophile** and attacks an **electrophilic center**.

LiAlH_4 reacts with a **carbonyl** by transferring a hydride to the carbonyl carbon, thereby increasing the number of C–H bonds, decreasing the number C–O bonds, and reducing the oxidation state of the carbonyl carbon. The ester in Compound 1 contains a carbonyl, and therefore will react with LiAlH_4 and form two alcohols, splitting the molecule in two. The amide also contains a carbonyl and will react with LiAlH_4 to yield an amine.

(Choices A and D) The carbon-carbon double bond and the aromatic ring are not electrophilic centers, and therefore will not react with the nucleophilic hydride ion of LiAlH_4 .

(Choice C) The proton of the alcohol functional group will undergo an acid-base reaction with a hydride ion, but the C–O bond of the alcohol will not be reduced by LiAlH_4 .

Educational objective:

Hydride reagents, such as LiAlH_4 , act as nucleophiles and attack electrophilic centers, most commonly carbonyl carbons ($\text{C}=\text{O}$). These reagents cause a reduction of the electrophile, decreasing the number of bonds to electronegative atoms and increasing the number of bonds to less electronegative atoms.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401155

System:
Functional Groups and Their Reactions

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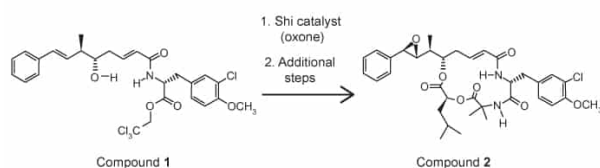


Figure 1 Shi epoxidation of Compound 1

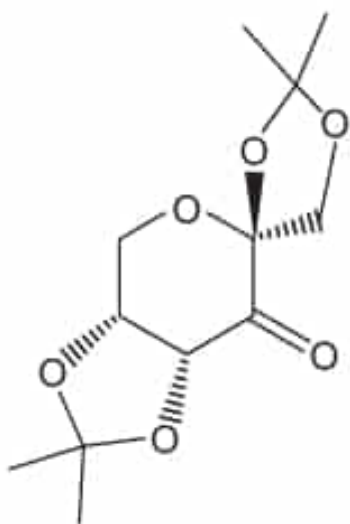


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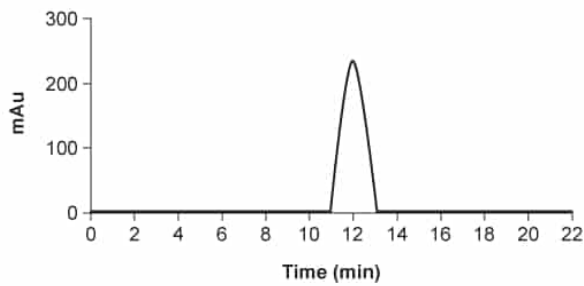
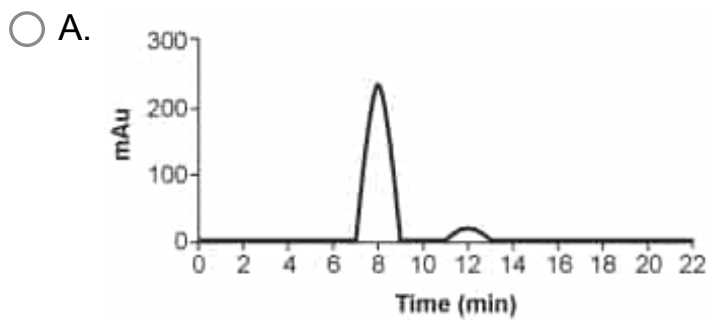
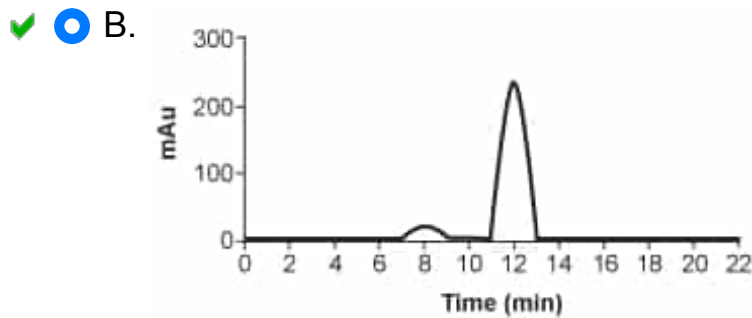


Figure 3 HPLC of epoxide standard

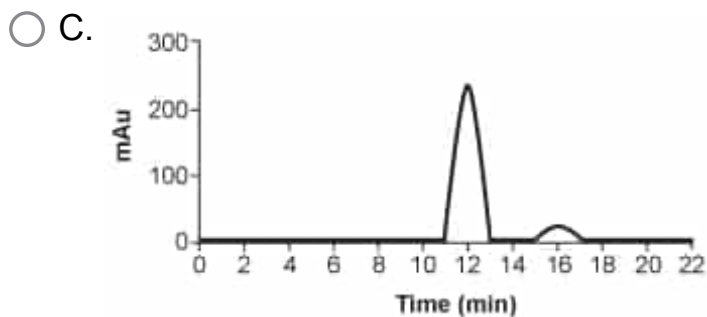
The HPLC analysis of the Shi epoxide revealed a 10:1 mixture of isomers in favor of the desired isomer. Based on the chromatogram shown in Figure 3, which of the following most likely depicts the chromatogram of the mixture?



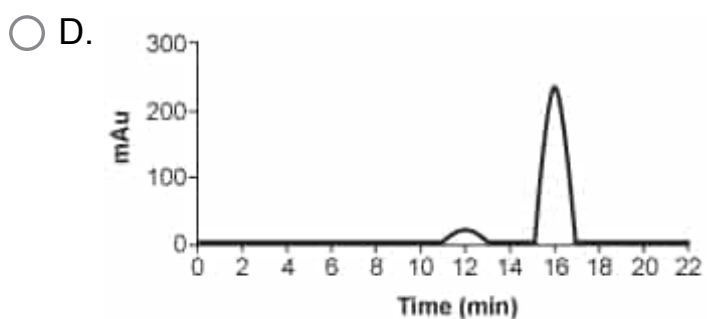
[5%]



[64%]



[25%]

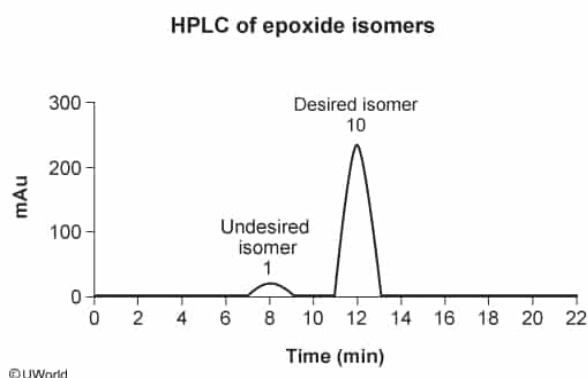


[5%]

Correct

64% answered correctly

Explanation:



High-performance liquid chromatography (HPLC) is a purification method that separates mixtures based on the component's **affinity** for the **stationary phase**. Compounds that have an affinity for a column interact with the stationary phase and are retained on the column for a longer time than those that do not have an affinity for the column. Chiral columns have a higher affinity for one configuration of a stereocenter than the other. A **standard** can be used to optimize conditions for the purification and to **predict** the **retention time** of a known compound.

Figure 3 shows that the standard of the desired isomer has a retention time of 12 minutes. The passage states that the desired isomer has a higher affinity for the chiral HPLC column than other isomers; therefore, it will be retained on the column longer than the undesired isomer, which has a lower affinity for the column. The question further states that the mixture is 10:1 in favor of the desired epoxide isomer, meaning that the peak of the desired isomer will be larger than the peak of the undesired isomer.

(Choice A) Although the desired isomer should elute at 12 min and the undesired isomer should elute earlier, the peak at 12 min should be larger because there is a 10:1 ratio of desired to undesired isomers.

(Choice C) This chromatogram correctly shows more of the isomer with a retention time of 12 min. However, the undesired isomer should elute before 12 min because it has a lower affinity for the column.

(Choice D) Based on Figure 3, the standard of the desired isomer has a retention time of 12 min. Therefore, the larger peak should elute at 12 min and the smaller peak at <12 min.

Educational objective:

High-performance liquid chromatography can be used to separate mixtures into their

constituents using a column that has an affinity for one of the compounds over another in the mixture. A standard shows the expected retention time of a given molecule.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401156

System:
Functional Groups and Their Reactions

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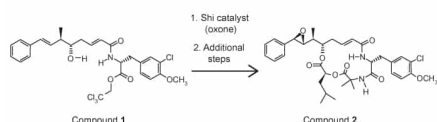


Figure 1 Shi epoxidation of Compound 1

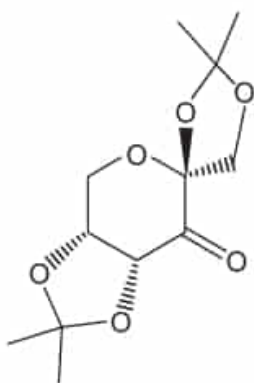


Figure 2 Structure of the Shi catalyst

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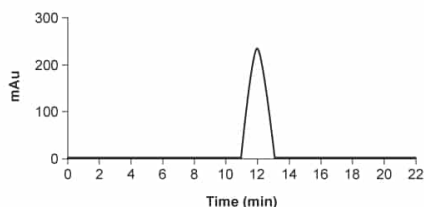


Figure 3 HPLC of epoxide standard

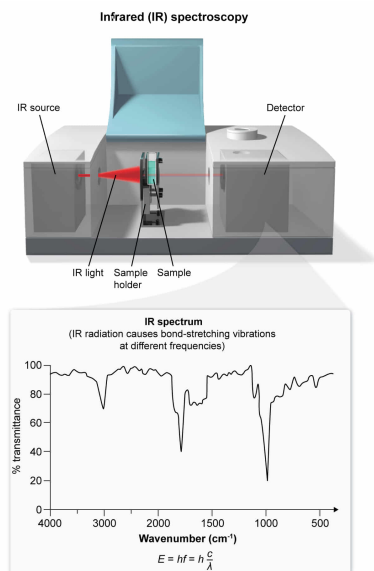
After isolation of Compound 1, an infrared spectrum was obtained. The spectrum gives all the following information EXCEPT:

- ☒ A. the spin-spin splitting of atoms in a compound. [79%]
- ☐ B. the signals corresponding to stretching vibrations and rotations. [3%]
- ☐ C. the amount of light absorbed at a certain frequency. [10%]
- ☐ D. the relative amount of energy needed to stretch a bond. [7%]

Correct

79% answered correctly

Explanation:



Infrared (IR) spectroscopy is a technique used to **identify functional groups** in a compound by analyzing the **types of bonds** present. The sample is irradiated with IR light, which interacts with the bonds in the molecule. The amount of light absorbed is recorded by the detector, and the data collected is displayed on the IR spectrum as **percent transmittance** (y-axis) versus **wavenumber** (x-axis). Transmittance is inversely related to absorbance (ie, if all the light is absorbed, there is 0% transmittance). The wavenumber (with units of cm⁻¹) is the IR frequency and is defined as the inverse of the wavelength. Because the wavenumber is related to wavelength, it is also directly related to frequency by the equation:

$$f = \frac{c}{\lambda}$$

where f is the frequency, c is the speed of light, and λ is the wavelength. Therefore, IR spectroscopy measures absorbance at different frequencies (**Choice C**), and frequency is directly related to energy (**Choice D**). The energy needed to stretch a bond at a certain wavelength can be calculated using:

$$E = hf = h \frac{c}{\lambda}$$

where E is energy and h is Planck's constant. IR radiation ranges from 700 nm to 1 mm and causes bond **stretching vibrations and rotations** at **different frequencies** (**Choice B**). **Functional groups** absorb different amounts of energy depending on the type of bond present. Therefore, the **intensity of the signals** from bond-stretching vibrations and rotations at a certain **frequency** depends on the amount of **energy absorbed**.

Spin-spin splitting of atoms is a phenomenon that occurs in nuclear magnetic resonance (NMR) spectroscopy. It revolves around how protons act in a magnetic field and the effect they have on the signals of neighboring protons.

Educational objective:

Infrared spectroscopy is a technique used to determine the functional groups present in a sample; the data collected are plotted as percent transmittance vs. wavenumber. The signals in the spectrum correspond to bond-stretching vibrations and rotations at a certain frequency, and the signal intensity is dependent on the amount of energy absorbed.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401157
System:
Functional Groups and Their Reactions